

Malay Tweets: Discovering Mental Health Situation during COVID-19 Pandemic in Malaysia

Ramadani Anwar Sabaruddin
Faculty of Computer Science and
Information Technology
Universiti Malaysia Sarawak
Sarawak, Malaysia
danisabaruddin98@gmail.com

Suhaila Saeed
Faculty of Computer Science and
Information Technology
Universiti Malaysia Sarawak
Sarawak, Malaysia
ssuhaila@unimas.my

Abstract— During the unprecedented of COVID-19 pandemic, numbers of research had been conducted on mental health in social media worldwide. Past research has shown interest in Twitter sentiment analysis by using keywords, geographical area, and range of ages. Up to the authors' analysis, there is no research conducted on mental health using keyword in the case of Malaysia. A Malay Tweet dataset was built for analysing mental health tweets during the first Movement Control Order period using unique keywords. Machine learning algorithms namely, Naïve Bayes classifier and Support Vector Machine were used to predict the sentiment of tweets. The classifiers were evaluated using 10-fold cross-validation, accuracy, precision, and F1-score. The data then visualized in charts and *WordCloud*. The results shows that Support Vector Machine performed better than Naïve Bayes classifier for both test set and 10-fold cross-validation in terms of performances in n-gram TF-IDF. The visualized data could provide insights to the authority pertaining the mental health issues, in which it relates to local news and situations during the periods.

Keywords— COVID-19, Social media NLP, Machine Learning

I. INTRODUCTION

COVID-19 disease originated from Wuhan, China. Ever since, people around the world are appalled since World Health Organization (WHO) declared the novel coronavirus spread as a global pandemic on 11 March 2020, as more than 118,000 of cases of the coronavirus disease in 110 countries and possibility arises for further global spread [8]. The Government of Malaysia has implemented the Movement Control Order (MCO) started from 18th March 2020. [3] stated that anxiety and fear are natural responses to the COVID-19 pandemic. During the MCO, new normal practices such as social distance and being quarantined may trigger different type of emotions such as anxiety, fear, frustration, and loneliness.

Social media such as Facebook and Twitter are the platforms for people to express themselves, especially during the pandemic as number of people are required to work from home, and students to attend classes, which can be depressive for some group of people. Numbers of previous research utilizing machine learning (ML) techniques to analyse text data from Internet on mental health problems has been identified such as in [10] and [2]. Analysis of Tweets related to the mental health issues is suggested as most studies are using keywords to crawl Twitter data such as [11] and [12].

There are significant numbers of studies regarding mental health in Twitter during COVID-19 pandemic, but only few that conducted in Malaysia context. For example, the impact

of COVID-19 pandemic on mental health in Malaysia by [22] as they only focused on explaining about how Malaysia handled the outbreak of COVID-19 pandemic such as implementation of MCO and how people are affected to it. [23] has conducted self-rating anxiety questionnaire to assess the anxiety of the students, but only focused among university students in Malaysia. In depth analysis is required as the previous work has only been limited to certain geographical area and range of ages. Objectives of the research are as follows: (1) to analyse on the mental health side of Malaysians in Twitter during COVID-19 pandemic using Twitter sentiment analysis and machine learning algorithms, (2) to obtain the best performance of modelling by comparing the machine learning algorithms, and (3) to visualize the mental health impact using visualization tool.

II. RELATED WORKS

The most common method of data acquisition are based on the survey, questionnaire and existing statistics. Few studies have been published on mental health effect to society. [27] has performed a research on their perspective of social media use and mental health during the pandemic event as moderator role of disaster stressor and mediator role of negative impacts by using online survey. [1] used survey or questionnaire to conduct research on human emotion as have acquired their data by handling questionnaire to Mechanical Turk workers located in Japan.

Whilst data crawling using Twitter API to acquire the Tweets data has been conducted in the digital age. [6] acquired their data by collecting crisis-related tweets that are posted during 19 different crises that took place from 2013 to 2015. [13] had collected their data by extracting a random sample of tweets in English language and its advanced search tool which produced a relevant subset of tweets that does not have any retweets from 14 January to 28 January 2020. *Tweepy* had been used by [26] in their paper regarding monitoring depression trend on Twitter.

Based on the reviews conducted, the general pipeline in the data preparation are: (1) data sampling, (2) data collection, and (3) data pre-processing. Two steps involved in the data analysis: (1) unsupervised ML, which is using the clustering algorithms, sentiment analysis and thematic qualitative analysis. In data sampling and collection steps, [25] had collected Twitter feeds by using a list of COVID-19 related hashtags as search terms and pre-processed the data. [25] and [13] were using *Python* for the pre-processing part, but *RStudio* was used in [13].

In terms of sentiment analysis, sentiment polarity explained emotions in term of deep-seated attractiveness and frigidness of a subject [13]. Sentiment analysis has also been used in [25] and [13], as they normalized the scores to detect the polarity of the tweets as *positive*, *negative*, or *neutral*.

Few works implemented *Linguistic Inquiry* and *Word Count*. As defined in [16], LIWC is a transparent text analysis program that counts words in psychologically meaningful categories. Based on [26]’s paper on monitoring depression trend on Twitter during the pandemic of COVID-19, they use LIWC to extract people’s psychological states by analysing their tweets contents.

From the aspects of ML, two ML algorithms were reviewed. Naïve Bayes (NB) classifier is a supervised ML classifier which uses method of statistics for solving classification problems [15]. The classifier works by searching properties, considering all the features are mutually independent. In [9], NB classifier is used after labelling the dataset. Support Vector Machine (SVM) is one of the well-known supervised ML algorithms to perform text classification. Kernel function in SVM is used to change the data into proposed form. As in [14], the SVM classifier was trained using unigram, bigram, trigram, and combination of these n-grams with labelled tweets.

To evaluate the ML performances, a confusion metric is the most common metrics to be applied. Precision, recall, and F1-score were used in [2] and [26]. To visualise the Tweets data, [1], [7] and [11] used *WordCloud*, which is a cluster of words, depicted in different sizes and colours, depending on its weightage and importance. Unlike [5] and [12], a traditional bar chart, pie chart and line chart used to visualize their data. These visualizations also help readers to understand more about their result, as the graph and chart are depicted in such various colours and style.

III. METHODOLOGY

A. Phase 1: Data Acquisition and Preparation

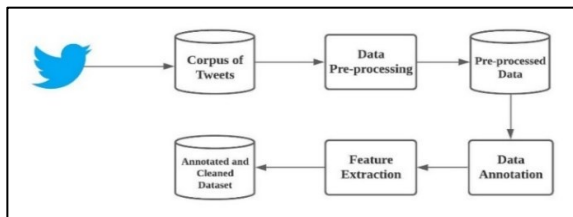


Fig. 1: Phase 1 Processes

1) Data Collection

There are 2549 tweets were collected using *Twint Python* library. The process was done by feeding the crawler with the six unique search strategy keywords related to mental health, which includes hashtags: *#KitaJagakita*, *#covid19*, *sedih* (*sad*), *marah* (*angry*), *kecewa* (*disappointed*) and *bunuh diri* (*suicide*). *#covid19* are globally used by Twitter users to address various kind of tweets genre pertaining to the COVID-19. On 19 March 2020, the doctors started the movement by using *#stayathome* and *#KitaJagaKita* in the social media, which resulted widely used by Malaysian users in Twitter [21]. The strings *sedih*, *marah*, *kecewa* and *bunuh diri*, were chosen in this work as these words are related to

the emotions felt by Malaysians in the early stage of the pandemic. The choice of these strings of words are identical of strings used by [24] with some alterations, as they had used human-mad disaster-related keywords such as anger, anxiety, and sadness.

Sixteen major cities around Malaysia were chosen as one of the requirements besides the mentioned six keywords. The major cities included *Johor Bahru*, *Alor Setar*, *Kota Bharu*, *Bandar Melaka*, *Seremban*, *Kuantan*, *Georgetown*, *Ipoh*, *Kangar*, *Kota Kinabalu*, *Kuching*, *Shah Alam*, *Kuala Lumpur*, *Victoria*, and *Putrajaya*. The period of date for the crawled tweets has been set from 18 March 2020 until 31 December 2020. Initially, the features collected by *Twint* were *id*, *conversation_id*, *created_at*, *date*, *time*, *timezone*, *user_id*, *username*, *name*, *place*, *tweet*, *language*, *mentions*, *urls*, *photos*, *replies_count*, *retweets_count*, *likes_count*, *hashtags*, *cashtags*, *link*, *retweet*, *quote_url*, *video*, *thumbnail*, *near*, *geo*, *source*, *user_rt_id*, *user_rt*, *retweet_id*, *reply_to*, *retweet_date*, *translate*, *trans_src*, and *trans_dest*. Since some of the features were not meaningful in this work, hence, the rest of the features were excluded in the crawling process. The remaining features that are useful for classification purposes are *id*, *user_id*, *username*, *tweet*, *date* and *near*.

2) Pre-processing Stage.

The crawled tweets consist of raw data that included links, emoji, and punctuations. During pre-processing by *Python* code, there are specific elements that are removed and replaced, such as removal of punctuations, numbers, special characters, emoji, removal of URLs and changing uppercase letters to lowercase. Human intervention is required to ensure no impurities left on the dataset. The intervention tasks are hashtag segmentation, removal of *id* and *user_id* columns and replacing them with *tweet_ID* and added two new columns of *keyword* and *month*.

3) Sentiment Annotation.

As NB and SVM are supervised learning approach, they required data annotation. Semi-automatic sentiment annotations were performed, whereby the tweets were annotated by both *Python* codes and human. First, tweets are automatically annotated by feeding each tweet to *Malaya library*. Probabilities of tweet sentiment were yielded, as the highest probability among three is the sentiment label for the tweet. Per author’s observation, data annotated by *Malaya library* were less accurate as it needs human intervention.

4) Feature Extraction.

TF-IDF vectorizer was implemented to convert the tweets to TF-IDF feature matrix from *sklearn Python* library. Next, word-level n-grams, namely, unigram and bigram were used to the Tweet.

B. Phase 2: Classification

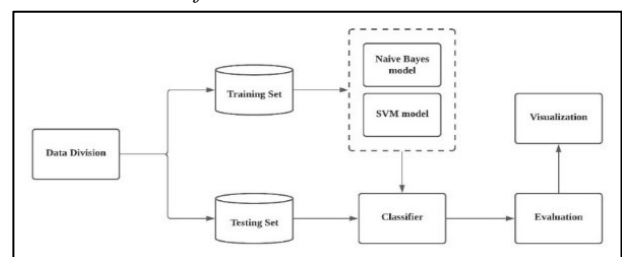


Fig. 2: Phase 2 Processes

The dataset was split into two sets which consists of 80 percent for training and 20 percent for testing, as [18] stated that 80:20 ratio is the most accurate when compared to other ratios (70:30, 50:50).

Naïve Bayes classifier is a supervised machine learning classifier which uses method of statistics for solving classification problems. The classifier works by searching properties, considering all the features are mutually independent. NB was chosen as its classifier yields good result in when it comes to text analysis. Support Vector Machine is one of the well-known supervised machine learning algorithms to perform text classification. SVM's main point is searching for a linear separator in the search space that can separate classes. Kernel function in SVM is used to change the data into proposed form. SVM is one of the effective learning algorithms for classifying text documents [19]. After building the classifier, the model was trained using the training set, and the test set was used to predict the outcome. The performance of the algorithms are evaluated by comparing 10-fold cross-validation with testing accuracy, calculating precision, recall, and F1-score for both unigram and bigram TF-IDF. 10-fold cross-validation was used in the experiment as it is a common practise in data mining and machine learning work [20], although the accuracy of $k=5$ and $k=10$ are relatively similar during the experiment.

IV. RESULTS

The result analysis of the project compared both ML classifiers. Firstly, scores among the algorithms were compared. For NB, there were slightly difference in test set score between unigram and bigram TF-IDF, as well as 10-fold cross-validation scores. Comparing the performance of test set and 10-fold cross-validation method, 10-fold cross-validation performed the highest score. The time taken calculated is the execution time for training. Table 1 depicts the performance evaluation result for both NB and SVM algorithms.

TABLE 1. PERFORMANCE EVALUATION RESULT OF CLASSIFICATION

	Naïve Bayes		Support Vector Machine	
	Unigram TF-IDF	Bigram TF-IDF	Unigram TF-IDF	Bigram TF-IDF
Test Set	0.75	0.80	0.85	0.84
10-fold CV	0.77	0.80	0.85	0.83
Time taken (s)	0	0	7	10

Based on Table 1, SVM performed the highest score in test set compared to NB. Also, 10-fold cross validation for SVM yielded higher score than NB. Notice that the time taken for both classifiers are significantly different, whereby NB executes better than SVM. This is because NB utilized probabilities which can be easily executed with such type of document, while SVM involved geometric calculation, which re-quires more time to classify the tweets.

The result demonstrated that NB classifier obtained 0.75 which means that in 75% of time the tweet sentiment was correctly predicted with test set, while using 10-fold cross-validation, the classifier obtained 0.85 which means that in 85% of time the tweet sentiment was correctly predicted. For

SVM classifier however, there were only a slight difference as when using test set SVM classified the sentiment of tweet 80% correctly, while scores 0.84 which means 84% of predicted sentiment of tweet was correct when using 10-fold cross-validation. In conclusion, both NB and SVM could achieve good predicting results in the sentiment analysis, especially SVM, as it has the highest score for both test set and 10-fold cross-validation. The score for SVM for both test set and 10-fold cross validation were the same, as the test set size might be equivalent to the average iterations of 10-fold cross validation. The test set and 10-fold cross-validation for NB obtain similar values of 0.80, which means 80% of prediction are correct. This was due to identical classification features for both test set and 10-fold cross validation set. For SVM, score of 0.84 was obtained in test set, while 0.83 was obtained in 10-fold cross-validation. Hence, SVM has the highest accuracy compared to NB, for both unigram TF-IDF and bigram TF-IDF. This result resembles the same output from [4], where they had compared various machine learning algorithms with different domain, as for Amazon, the SVM score for unigram and bigrams is the same.

Precision, recall and F1-score were chosen for performance metric as in [26] and [2]. The results were divided into three sentiment type to be analysed among two classifiers, which are negative sentiment, neutral sentiment, and positive sentiment, comparing between NB and SVM classifiers, as follows: -

A. Negative Sentiment.

It could be deduced that NB achieved perfect score for precision, compared to SVM with 0.01 score deficit, as it shown that the predictions were all precise in NB and SVM. SVM scored higher compared to NB with 0.05 deficit in terms of recall as of all negative tweets in SVM were classified, 84% of them were correctly predicted. Comparing both precision and recall, the F1-score for SVM was higher with 0.03 difference. Table 2 shows the overall negative sentiment performance scores for both compared ML classifiers.

TABLE 2. NEGATIVE SENTIMENT PERFORMANCE SCORES

	Naïve Bayes		Support Vector Machine	
	Unigram TF-IDF	Bigram TF-IDF	Unigram TF-IDF	Bigram TF-IDF
Precision	1.00		0.99	
Recall	0.79		0.84	
f1-Score	0.88		0.92	

B. Neutral Sentiment.

There was a significant difference between precision score of NB and SVM classifiers for neutral sentiment. The 0.19 score deficit between the two ML classifiers shown that SVM was better in prediction neutral sentiment, as it scored 0.44. However, recall score for NB was slightly higher than SVM, with 0.06 deficit, as it depicted that NB was better in classifying the neutral sentiment, considering of all the neutral sentiment in the dataset. Taking account of precision and recall for both ML algorithms, SVM scored higher with 0.18 difference between them. Table 3 shows the overall neutral sentiment performance score for both NB and SVM.

	Naïve Bayes		Support Vector Machine	
	<i>Unigram TF-IDF</i>	<i>Bigram TF- IDF</i>	<i>Unigram TF-IDF</i>	<i>Bigram TF- IDF</i>
Precision	0.25		0.44	
Recall	0.93		0.87	
f1-Score	0.40		0.58	

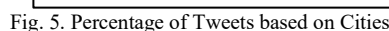
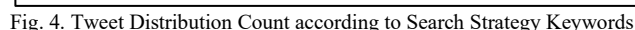
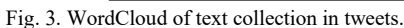
C. Positive Sentiment.

For positive sentiment, unusual cases had occurred, where NB has zero score for precision, recall and F1-score. Although NB yielded significant scores for accuracy for both test set and 10-fold cross validation, precision score for NB was zero due to no precision in predicting positive sentiments of all correctly predicted positive sentiments. Same goes to recall, whereby might be due to zero denominator of the equation to quantify the recall, thus the zero score for F1-score which taking account of precision and recall. For SVM however, 23% prediction of positive sentiment, which means that 23% of all the predictions were precise. SVM recorded perfect score for recall, which means that 100% of the positive sentiment prediction was correct of all the positive sentiment in the dataset. Hence, F1-score for SVM was 0.38 significantly higher than NB. Table 4 shows the positive sentiment performance scores for NB and SVM.

	Naïve Bayes		Support Vector Machine	
	<i>Unigram TF-IDF</i>	<i>Bigram TF-IDF</i>	<i>Unigram TF-IDF</i>	<i>Bigram TF-IDF</i>
Precision	0.00		0.23	
Recall	0.00		1.00	
f1-Score	0.00		0.38	

Upon data collection, a total of 2548 tweets have been crawled, as 2252 tweets are in the final dataset after pre-processing and annotation. Figure 5 shows the count of crawled tweets based on the search strategy keywords.

As illustrated in Figure 4, users use *bunuh diri* for 821 times, which prevailed as the highest usage among all the keywords, followed by *kecewa* - 522 times, *marah* - 479 times, *#KitaJagaKita* - 162 times, *kecewa* - 152 times, and *#covid19* - 116 times. The highest numbers of *bunuh diri* tweets during the first MCO could be related to the number of Malaysians committed suicide. As such, 266 suicide cases were reported from 18 March to 30 October 2020 as stated in a news article by *Kosmos Digital*¹.



Based on Figure 5, 25.9% of the mental health related tweets were came from Shah Alam with a total of 577 tweets, as Kuala Lumpur recorded the second highest with 451 tweets, followed by Putrajaya with 432 tweets. The huge number of these cities are likely biased as some of the cities were not recognized by the tweet crawler. Next, the tweet count for each keyword by months were analysed. Figure 5 visualized the number of tweets for each keyword in *March 2020*.

As depicted in Figure 6, usage of *bunuh diri* were the highest started April to December when compared to other unique keywords used. Based on the news article in *Astro Awani*², the rate of suicide increased from 2019 to November 2020 by 20%. The demographic of the statistics shows that Malaysians ranged 19-30 years old and 31-45 years old has significantly increased compared to 2019. The second highest usage of word is *sedih* (sad). Based on the crawled tweets, there are Malaysians felt sad of losing someone they love due to this pandemic..

Based on Figure 7, it can be concluded that *bunuh diri* has the highest count among all the words in the column. This shows that suicide is one of the most concerning case in mental health issues, as *bunuh diri* has the biggest size in the *WordCloud*. The second highest are *diri sendiri* which appeared 553 times in tweet column, as the tweets are about Malaysians’

¹ <https://www.kosmo.com.my/2020/11/18/266-bunuh-diri-ketika-pkp-dilaksana/>

² <https://www.astroawani.com/berita-malaysia/kes-bunuh-diri-naik-mendadak-melonjak-20-peratus-berbanding-tahun-lepas-272931>

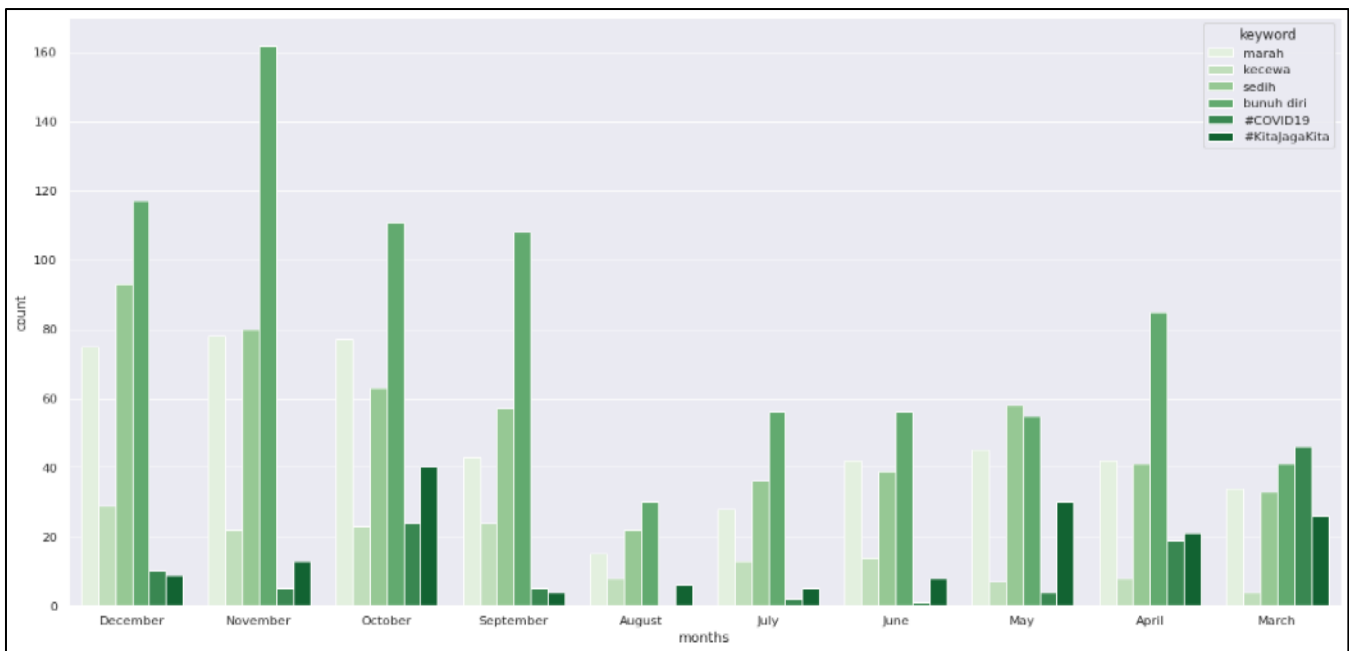


Fig. 6. Tweet Distribution count based on Keywords by Months

Based on Figure 7, it can be concluded that *bunuh diri* has the highest count among all the words in the column. This shows that suicide is one of the most concerning case in mental health issues, as *bunuh diri* has the biggest size in the *WordCloud*. The second highest are *diri sendiri* which appeared 553 times in tweet column, as the tweets are about Malaysians' struggles in handling their emotions in this very early stage of MCO, which still does not end until December 2020. Also, based on the *WordCloud* and crawled tweets, it can be deduced that Twitter users are very sad and angry regarding the commencement of MCO, as unemployment rate increases and some of the users experienced depression because they cannot be with their loved one during this trying time. Referring to *Berita Harian*³ article dated 31 October 2020, more than 800,000 Malaysian had lost their job due to COVID-19 pandemic in Malaysia as the government is giving their best solution to curb the spread of the infection and help those who are severely affected.

VI. CONCLUSIONS

To conclude, SVM outperformed NB for both unigram TF-IDF and bigram TF-IDF in terms of the accuracy. This shows that SVM could predict more precise sentiment classification. All significant data had been visualized as they were analysed to show the trend and statistics that are related to mental health issue happened in Malaysia during COVID-19. Besides, the contribution from this work could address the issues facing by the society whereby the government of Malaysia should take a more serious action upon mental health issues in Malaysia.

Few limitations in the work, namely, the capabilities of *Twint Python* library and lack of Malay annotated wordlist. The downside of *Twint Python* library is the tweet crawler has longitude and latitude parameters that can be used to crawl all the tweets in certain geolocations. However, the library is less effective for Malaysia context. Lastly, the lack of Malay

annotated wordlist in *Malaya Python* library resulted more human intervention to correct the auto generated sentiment results. Indirectly, the task is time consuming.

VII. ACKNOWLEDGEMENT

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³<https://www.bharian.com.my/berita/nasional/2020/10/748412/800000-hilang-kerja-sejak-covid-19-melanda>

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